Astronomy - Graduate Level Assessment 5 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Approximately how far away is the Andromeda Galaxy?
   (a) 1.7 million light years
   (b) 2.1 million light years
   (c) 2.5 million light years
   (d) 3.2 million light years
2. What Mars mission will be landing on May 25 2008 and will dig a trench into (hopefully) ice-rich soil?
   (a) ExoMars
   (b) Mars Exploration Rovers
   (c) Mars Science Laboratory
   (d) Phoenix Mars Lander
3. Find the best approximation for the surface temperature of the Sun:
   (a) 6000 K
   (b) 7000 K
   (c) 9000 K
   (d) 13000 K
4. Calculate the ratio of the solar radiation flux on Mercury's surface for perihelion (0.304 AU) versus aphelion (0.456 AU).
   (a) 4:1
   (b) 1:2
   (c) 6:5
   (d) 9:4
5. Why are Cepheid stars relevant for astronomers?
   (a) To measure interstellar mass.
   (b) To measure galactic distances.
   (c) To measure galactic energy-density.
   (d) To measure interstellar density.

True / False (10 marks)

Answer True or False

6. True or False: Titan is the only outer solar system moon that has liquid water on its surface, indicating potential habitability.
7. True or False: Planetary rings can be formed from the dismantling of small moons by impacts over time.
8. True or False: The black hole at the center of the Milky Way is referred to as Alsephina A*.
9. True or False: Kirkwood gaps occur in the main asteroid belt at positions where asteroids orbit with a period twice that of Mars.
10. True or False: The belts and zones of Jupiter are alternating bands of rising and falling air at different latitudes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the significance of hydrogen as the most abundant element among the Jovian planets and analyze its implications for their formation and atmospheric characteristics.
12. Calculate the mass of a spherical moon with a diameter of 200 km composed entirely of water ice, and analyze the ratio of this mass to that of Saturn's rings, which is 2x10ˆ19 kg. Discuss the implications of this ratio in the context of celestial body composition and formation.

End of Paper